7 One possible nuclear reaction that takes place is

$$92235U + 01n \longrightarrow 4295Mo + 57139La + 2\,01n + 7 - 10e$$

Some masses are shown in Fig. 7.1.

	mass / u
proton ($11p$)	1.007 276
neutron ($01n$)	1.008 665
uranium-235 nucleus ($92235U$)	235.043 930

Fig. 7.1

(a) **(i)** Show that the binding energy per nucleon of a uranium-235 nucleus is 7.409 MeV.

[3]

(ii) The nucleon number of an isotope of the element rutherfordium is 267.

State whether the binding energy per nucleon of this isotope will be greater than, equal to or less than the binding energy per nucleon of uranium-235.

..... [1]

(b) The binding energy per nucleon of molybdenum-95 ($4295Mo$) and lanthanum-139 ($57139La$) are 8.443 MeV and 8.189 MeV respectively.

Calculate the total energy, in MeV, released in this nuclear reaction.

energy = ... MeV [2]

(c) The nuclei in 1.2×10^{-7} mol of uranium-235 all undergo this reaction in a time of 25 ms.

Calculate the average power release during the time of 25 ms.

power = ... W [3]

[Total: 9]